

CITY A

Structure A: direct

Background: Residents receive campaign developments through synchronized citywide alerts. Reactions occur mostly **during the same week as the news**.

week	net news	support
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1	4.0	53.1
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2	12.0	59.3
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3	0.0	50.0
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4	0.0	50.0
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no news:
back at 50

HIDDEN RULE (never shown to the model)

$$y_t = 50 + g x_t$$

$g = 0.771$; only this week's news counts

CITY B

Structure B: persistent

Background: Campaign developments usually spread through neighborhood conversations and local organizations. Reactions may **build gradually and persist** after the original story has faded.

week	net news	support
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1	4.0	51.9
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2	12.0	57.6
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3	0.0	56.5
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4	0.0	54.4
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no news, yet
still above 54

HIDDEN RULE (never shown to the model)

$$m_t = \rho m_{t-1} + g x_t \quad (m: \text{hidden local state})$$

$$y_t - 50 = \phi (y_{t-1} - 50) + \lambda m_t$$

$$g = 0.666, \rho = 0.551, \phi = 0.354, \lambda = 0.732$$

CITY C No observations from City C are available yet.

Apply net campaign news of +12.0 in the next week, then set it to 0.
Forecast City C's candidate support 3 weeks later.

If City C works like City A: predict 50.0

If City C works like City B: predict 53.6